Revealing Key Details to See Differences: A Novel Prototypical Perspective for Skeleton-based Action Recognition

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Abstract

Skeleton-based action recognition faces a fundamental challenge: distinguishing visually similar actions that differ only in subtle joint-motion details. Existing graph convolutional networks (GCNs) typically learn coarse motion patterns over fixed or adaptively learned topologies, failing to capture the fine-grained discriminative cues confined to specific body parts. In this paper, we propose ***ProtoGCN****, a novel framework that addresses this limitation through a prototypical perspective on motion topology. ProtoGCN introduces three complementary components: (1) a* ***Motion Topology Enhancement (MTE)*** *module that enriches each GCN layer with intra-sample and inter-sample topology signals, capturing both self-correlations and pairwise differences among joints; (2) a* ***Prototype Reconstruction Network (PRN)****, a learnable memory of motion prototypes with soft attention-based addressing that reconstructs topology features into discriminative representations; and (3) a* ***Class-Specific Contrastive Learning (CSCL)*** *objective that employs a momentum-updated class-center memory bank to enforce intra-class compactness and inter-class dispersion. The combination of prototype-guided reconstruction and contrastive regularization enables ProtoGCN to amplify subtle joint-motion differences that distinguish confusable actions. Extensive experiments on four benchmarks demonstrate the effectiveness of our approach: ProtoGCN achieves state-of-the-art results of 93.8%/97.8% on NTU RGB+D X-Sub/X-View, 90.9%/92.2% on NTU RGB+D 120 X-Sub/X-Set, 51.9% Top-1 on Kinetics-Skeleton, and 95.9% on FineGYM, with the most significant gains on the hardest action classes.*

1. Introduction

Human action recognition is a cornerstone of computer vision with broad applications in surveillance, human-computer interaction, and sports analytics. Among the modalities available, skeleton data has attracted considerable attention due to its compact representation, robustness to appearance and lighting variations, and relative ease of extraction via pose estimation systems [1, 42]. Skeleton-based action recognition has consequently become an active research field, with graph convolutional networks (GCNs) [3, 24, 37] emerging as the dominant framework for modeling the spatial and temporal relationships encoded in joint sequences.

Despite significant progress, a key challenge persists: *distinguishing similar actions that share overlapping coarse motion patterns but differ in fine-grained joint-level details*. Actions such as "Writing" and "Typing on a Keyboard" (Figure 3) involve similar gross upper-body motions, and purely skeletal data lacks the appearance cues that differentiate them in RGB modalities. The discriminative information is concentrated in specific joints and their pairwise relationships, yet standard GCN approaches tend to learn smooth, globally distributed representations that average out these subtle distinctions.

The root of this problem lies in how existing methods construct graph topologies. Pioneering work [37] uses fixed, anatomically-defined adjacency matrices, capturing only physical joint connectivity. Subsequent adaptive topology methods [3, 20, 24] learn data-driven adjacency weights, but these are typically shared across samples or computed from local features of a single sample in isolation, failing to highlight what makes one action *different* from another. The result is representations that are compact within each class but not maximally separated across similar classes.

In this paper, we draw inspiration from prototype-based learning [28], which represents classes through reusable exemplars and has shown strong performance in few-shot recognition. We hypothesize that action dynamics can be decomposed as combinations of *motion prototypes*—recurring joint-relationship patterns shared across actions—and that reconstructing sample features from these prototypes will expose their discriminative structure. Reconstructed features naturally emphasize which prototypes are activated and how strongly, providing a richer signal for distinguishing confusable actions.

We propose **ProtoGCN**, which integrates prototype-based feature reconstruction into the GCN architecture through three novel components:

Motion Topology Enhancement (MTE). Standard adap-

tive topologies capture within-sample feature correlations. MTE augments each GCN layer with two complementary topology signals: an *intra-sample* graph that captures self-correlations among joints through outer products of temporally-pooled features, and an *inter-sample* graph that captures pairwise differences between joints, highlighting relative motion patterns. Together, these enriched topologies provide a richer input for prototype retrieval.

Prototype Reconstruction Network (PRN). At the output of the GCN backbone, PRN maintains a learnable memory matrix of n_{pro} motion prototypes. Given the topology features of a sample, it computes a soft attention distribution over the prototype memory using a query transformation, then reconstructs the features as a weighted combination of prototypes. This reconstruction reveals which motion patterns are present in the sample and to what degree.

Class-Specific Contrastive Learning (CSCL). To enforce that reconstructed representations cluster tightly by class, we introduce a contrastive objective operating on momentum-updated class-center embeddings. CSCL computes feature similarities against a running memory bank of class prototypes and optimizes for pulling same-class features together while pushing different-class features apart.

We train ProtoGCN with a combined objective of cross-entropy and CSCL losses, with a scalar weight λ controlling the contrastive regularization strength. We evaluate on four benchmarks: NTU RGB+D [22], NTU RGB+D 120 [19], Kinetics-Skeleton [13], and FineGYM [23]. ProtoGCN achieves new state-of-the-art results on all four, with particularly pronounced gains on the most challenging classes—those most susceptible to confusion—demonstrating that our prototype-based reconstruction specifically addresses the fine-grained discrimination problem.

Our main contributions are:

- We propose **ProtoGCN**, a prototype-based GCN that reconstructs motion topology features from a learnable prototype memory to expose fine-grained discriminative joint dynamics.
- We design a **Motion Topology Enhancement (MTE)** module that enriches GCN layers with intra-sample self-correlation ($\mathbf{A}_{\text{intra}}$) and inter-sample difference ($\mathbf{A}_{\text{inter}}$) topology signals.
- We introduce a **Prototype Reconstruction Network (PRN)** with soft attention-based memory addressing for topology-aware feature reconstruction.
- We devise a **Class-Specific Contrastive Learning (CSCL)** objective with a momentum memory bank to enforce class-level compactness and dispersion.
- We achieve state-of-the-art results on NTU RGB+D 60/120, Kinetics-Skeleton, and FineGYM, with thorough ablations demonstrating the contribution of each component.

2. Related Work

Skeleton-based Action Recognition. Early approaches to skeleton-based action recognition relied on recurrent neural networks [8, 25, 26, 29] and convolutional networks [30] that process joint sequences as structured arrays. The introduction of ST-GCN [37] fundamentally transformed the field by formulating the skeleton as a graph and applying graph convolutions over spatially connected joints and temporally adjacent frames. This framework naturally encodes the kinematic structure of the human body and has inspired a large family of GCN-based methods.

Subsequent work focused on learning more expressive graph topologies. 2s-AGCN [24] proposed adaptive topology learning, adding data-driven edge weights to the fixed anatomical graph. CTR-GCN [3] learned channel-wise topology modulations to capture fine-grained joint interactions. MS-G3D [20] decomposed spatial-temporal dependencies through disentangled multi-scale graph convolutions. DC-GCN+ADG [5] introduced decoupled graph convolutions with aggregated decoupled graphs. More recent methods include HD-GCN [16], which exploits hierarchical decomposition of joints, BlockGCN [45], which decomposes the skeleton into body-part blocks, and DS-GCN [34], which captures dynamic spatial-temporal relationships. PYSKL [9] provides a comprehensive framework for skeleton-based recognition with strong baselines. Our work builds on these foundations by introducing a prototype-based reconstruction module that post-processes GCN topology features to amplify fine-grained discriminative patterns.

Graph Contrastive Learning for Skeletons. Contrastive learning has been applied to skeleton-based recognition to learn view-invariant representations. SkeletonGCL [12] introduces graph-level contrastive learning with skeleton-specific augmentation strategies. FR-Head [44] employs frequency-aware contrastive regularization to separate fine-grained motion features. InfoGCN [6] uses mutual information principles to encode context-dependent joint correlations. Our CSCL component differs in operating on *class-specific* prototype centers updated with a momentum mechanism, enabling contrastive regularization that directly targets inter-class confusion.

Prototype Learning. Prototype-based approaches [28] represent categories by their prototype embeddings and classify new instances by nearest-prototype matching. Hopfield networks and their modern dense versions [7, 15, 21] provide a theoretical foundation for associative memory with large capacity. Memory-augmented networks have been applied to few-shot learning [31, 39, 40] and dense prediction [38]. In NLP and vision, memory-based modules with attention addressing have shown strong performance for capturing compositional patterns [41]. We adapt these ideas to skeleton-based recognition: our PRN uses soft at-

tention to retrieve and combine motion prototypes, reconstructing topology features that highlight which recurring motion patterns are present in each sample.

Fine-Grained Action Recognition. Fine-grained recognition, where categories share similar global appearances but differ in subtle details, is a major challenge [11, 27]. For RGB-based methods, this has motivated part-based attention and region-pooling approaches. For skeleton-based recognition, the problem is particularly acute because skeletal data discards appearance entirely, leaving only structural and motion cues. FineGYM [23] is a challenging benchmark specifically designed to evaluate fine-grained discrimination among gymnastics routines. Methods like PoseConv3D [10] and SkeletonMAE [36] have advanced performance on this benchmark. ProtoGCN achieves a new state-of-the-art on FineGYM by specifically targeting the discriminative joint dynamics that distinguish visually similar routines.

3. Method

3.1. Problem Setup and Overview

Let $\mathbf{x} \in \mathbb{R}^{N\times M\times T\times V\times C}$ denote a batch of skeleton sequences, where N is the batch size, M is the number of persons (typically $M = 2$), T is the number of frames, V is the number of joints (e.g., $V = 25$ for NTU), and $C = 3$ is the coordinate dimension. The goal is to assign each sequence a class label from K action categories.

ProtoGCN processes each skeleton sequence through a GCN backbone augmented with Motion Topology Enhancement (MTE), then applies a Prototype Reconstruction Network (PRN) to the extracted topology features. The reconstructed topology features are fed to a Class-Specific Contrastive Learning (CSCL) loss during training, while the full backbone output is used for classification. The overall architecture is illustrated in Figure 1.

3.2. GCN Backbone with Motion Topology Enhancement

Our backbone follows the design of prior GCN-based methods [3, 9], consisting of 10 stacked GCN-TCN blocks that progressively expand the channel dimension from 96 to 384. We apply batch normalization on the input coordinates, normalized per vertex and channel.

Each GCN layer receives feature maps $\mathbf{x} \in \mathbb{R}^{N\times C\times T\times V}$ and an adjacency matrix, producing enhanced features. The key novelty is the **Motion Topology Enhancement (MTE)** module embedded within each GCN layer, which computes two complementary dynamic topologies.

Inter-sample Topology ($\mathbf{A}_{\text{inter}}$). To capture the relative motion differences between joints, we project the input features into a mid-channel space and temporally pool:

$$\mathbf{h}_1 = \frac{1}{T}\sum_t \text{Conv}_1(\mathbf{x})[\cdot, t, :], \quad \mathbf{h}_2 = \frac{1}{T}\sum_t \text{Conv}_2(\mathbf{x})[\cdot, t, :] \tag{1}$$

where both $\mathbf{h}_1, \mathbf{h}_2 \in \mathbb{R}^{N\times K\times C'\times V}$ with K graph subsets and C' mid-channels. The inter-sample topology is then:

$$\mathbf{A}_{\text{inter}}^{(i,j)} = \alpha \cdot \tanh(\mathbf{h}_1[\cdot,\cdot,\cdot,i] - \mathbf{h}_2[\cdot,\cdot,\cdot,j]) \tag{2}$$

for each joint pair (i, j), where α is a learnable scalar. The difference operation explicitly encodes how joint i differs from joint j in the feature space, making the topology sensitive to relative dynamics rather than absolute positions.

Intra-sample Topology ($\mathbf{A}_{\text{intra}}$). To capture pairwise self-correlations within the sample, we compute an outer product over joints:

$$\mathbf{A}_{\text{intra}}^{(i,j)} = \beta \cdot \text{softmax}\big(\mathbf{h}_1[\cdot,\cdot,\cdot,i]^\top \mathbf{h}_2[\cdot,\cdot,\cdot,j]\big) \tag{3}$$

where β is a learnable scalar. The softmax normalization ensures the intra-sample topology sums to 1 along each row, acting as an attention map over joint pairs.

Enhanced Adjacency. The final adjacency used in graph convolution is:

$$\mathbf{A} = \mathbf{A}_{\text{static}} + \mathbf{A}_{\text{inter}} + \mathbf{A}_{\text{intra}} \tag{4}$$

where $\mathbf{A}_{\text{static}}$ is the learnable baseline adjacency. The feature aggregation then proceeds as:

$$\mathbf{x}' = \mathbf{W}_{\text{post}}\left(\sum_k \hat{\mathbf{x}}_k \mathbf{A}_k\right) \tag{5}$$

where $\hat{\mathbf{x}}_k$ is the pre-projected feature for subset k, and $\mathbf{W}_{\text{post}}$ is the post-projection weight. The output of the GCN layer includes both the enhanced feature map and the combined topology graph $\mathbf{G} = \mathbf{A}_{\text{inter}} + \mathbf{A}_{\text{intra}}$, which is used downstream by the PRN.

3.3. Prototype Reconstruction Network

After passing through all GCN stages, we extract the topology graph $\mathbf{G} \in \mathbb{R}^{N\times C\times V\times V}$ from the final GCN layer. This topology encodes the learned joint-relationship patterns for each sample. The **Prototype Reconstruction Network (PRN)** transforms these topology features into a discriminative reconstructed representation by leveraging a learnable prototype memory.

Formally, we reshape the topology graph to $\mathbf{G} \in \mathbb{R}^{N\times C\times (V\cdot V)}$ and process each spatial position (v_i, v_j) independently. For a single sample, we have $\mathbf{g} \in \mathbb{R}^{V^2\times C}$ (transposed), and the PRN computes:

$$\mathbf{q} = \text{softmax}(\mathbf{W}_Q\, \mathbf{g}) \tag{6}$$

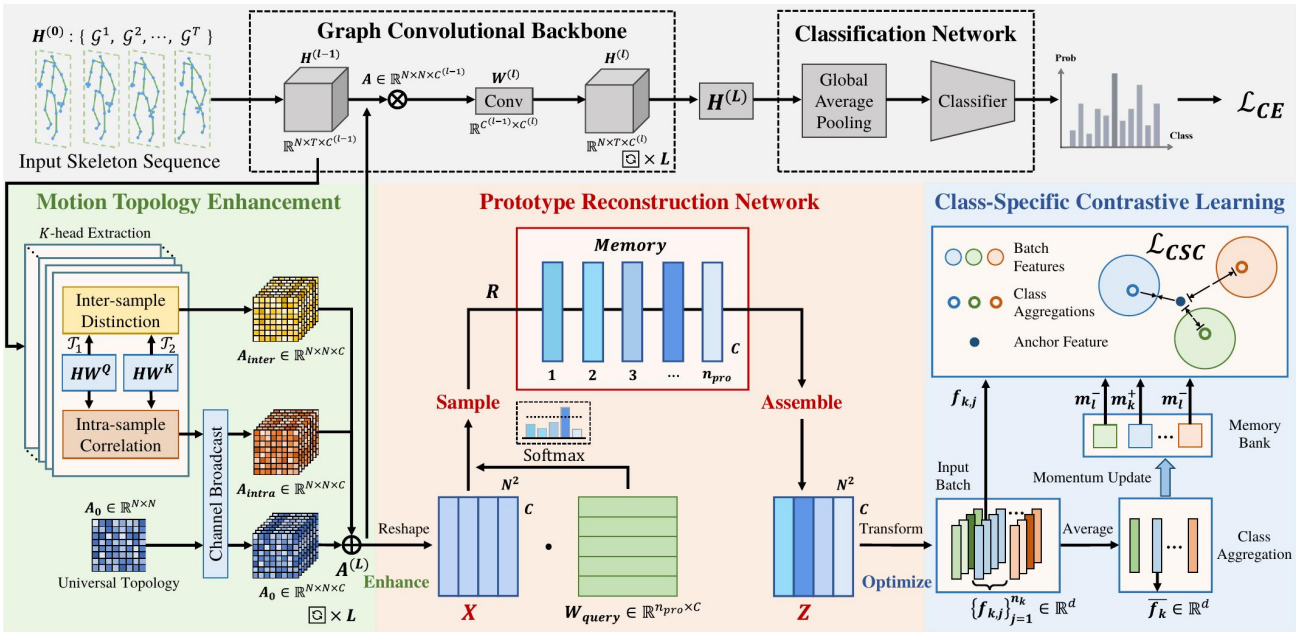

Figure 1. The overall architecture of ProtoGCN. Skeleton sequences are processed by a GCN backbone with Motion Topology Enhancement (MTE) at each layer, which computes intra-sample ($\mathbf{A}_{\text{intra}}$) and inter-sample ($\mathbf{A}_{\text{inter}}$) topology signals. The topology graph from the final GCN stage is fed to the Prototype Reconstruction Network (PRN), which reconstructs it through soft attention over a learnable prototype memory. The reconstructed topology features are used by the Class-Specific Contrastive Learning (CSCL) loss during training, while the GCN output features are used for classification.

$$\mathbf{z} = \mathbf{W}_M \, \mathbf{q} \tag{7}$$

where $\mathbf{W}_Q \in \mathbb{R}^{n_{\text{pro}} \times C}$ is the query matrix and $\mathbf{W}_M \in \mathbb{R}^{C \times n_{\text{pro}}}$ is the prototype memory matrix, with n_{pro} the number of prototypes. The softmax over the query output produces attention weights over the n_{pro} prototypes; $\mathbf{W}_M$ then retrieves a weighted combination of stored prototype patterns. A dropout layer is applied to the output for regularization.

The reconstructed representation $\mathbf{z} \in \mathbb{R}^{V^2 \times C}$ is reassembled into $\mathbb{R}^{C \times V \times V}$, processed by a post-convolution and batch normalization, and reduced to $\mathbb{R}^{V^2}$ by channel-mean pooling. This compact topology descriptor is passed to the CSCL loss.

The prototype memory $\mathbf{W}_M$ is learned end-to-end via backpropagation. The softmax attention ensures that each topology location is explained as a convex combination of prototypes, forcing the network to encode motion patterns in terms of reusable, discrete building blocks. This bottleneck structure encourages the PRN to discover and emphasize the prototype activations that best distinguish the current sample from potential confusers.

3.4. Class-Specific Contrastive Learning

To directly address inter-class confusion, we introduce a **Class-Specific Contrastive Learning (CSCL)** objective that operates on the reconstructed topology representations. CSCL maintains a momentum-updated memory bank $\mathbf{F} \in \mathbb{R}^{D \times K}$ of per-class center embeddings, where D is the projected feature dimension and K is the number of classes.

Feature Projection. The reconstructed topology descriptor is projected to a D-dimensional space via a linear layer: $\mathbf{f} = \text{FC}(\mathbf{z}_{\text{pool}}) \in \mathbb{R}^{N \times D}$.

Momentum Memory Update. For each training step, we identify correctly classified samples using a prediction mask $\mathbf{m} \in \{0, 1\}^{N \times K}$ that is non-zero only for samples whose predicted class matches the ground-truth class:

$$m_{n,k} = \mathbb{1}[\hat{y}_n = y_n = k] \cdot \mathbb{1}[\text{softmax}(\ell_n)_k > \tau] \tag{8}$$

where $\hat{y}_n$ is the predicted class, y_n is the ground truth, ℓ_n is the classification logit, and τ is a threshold. The class center for each class k is updated as:

$$\mathbf{f}_k \leftarrow \mu \, \mathbf{f}_k + (1 - \mu) \cdot \frac{\sum_n m_{n,k} \mathbf{f}_n}{\sum_n m_{n,k} + \epsilon} \tag{9}$$

with momentum μ and small constant ϵ. Classes without positive samples in the current batch retain their previous center estimate.

Contrastive Score and Loss. The contrastive similarity between a sample feature $\mathbf{f}_n$ and the class center $\mathbf{f}_k$ is com-

puted with cosine similarity scaled by temperature τ_c:

$$s_{n,k} = \frac{\hat{\mathbf{f}}_n \cdot \hat{\mathbf{f}}_k}{\tau_c} \tag{10}$$

where $\hat{\cdot}$ denotes ℓ_2 normalization. The CSCL loss is a cross-entropy over these similarity scores:

$$\mathcal{L}_{\text{CSCL}} = -\frac{1}{N}\sum_{n=1}^{N} \log \frac{\exp(s_{n,y_n})}{\sum_{k=1}^{K}\exp(s_{n,k})} \tag{11}$$

This loss pulls each sample's representation toward its class center and pushes it away from all other class centers, enforcing class-level compactness and inter-class dispersion specifically in the space of topology features.

3.5. Training Objective

The total training loss combines cross-entropy classification and CSCL:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{CSCL}} \tag{12}$$

where λ controls the strength of the contrastive regularization. We tune λ and n_{pro} through ablation experiments (Section 4.4).

4. Experiments

4.1. Datasets and Evaluation

NTU RGB+D (NTU-60) [22] contains 56,880 video clips of 60 action classes collected from 40 subjects. We report accuracy under two standard protocols: Cross-Subject (X-Sub), where training and test sets use different subjects, and Cross-View (X-View), where training uses cameras 2 and 3 while testing uses camera 1.

NTU RGB+D 120 (NTU-120) [19] extends NTU-60 to 114,480 clips of 120 classes from 106 subjects. We evaluate under Cross-Subject (X-Sub) and Cross-Setup (X-Set) protocols.

Kinetics-Skeleton [13] is derived from the Kinetics-400 video dataset by applying a pose estimation pipeline [1] to extract 2D skeleton sequences for 300K clips across 400 classes. We report Top-1 and Top-5 accuracy.

FineGYM [23] is a fine-grained action recognition benchmark capturing gymnastic routines at multiple granularities. We use the skeleton modality and report mean class Top-1 accuracy on the gym99 split, following the protocol of [9].

4.2. Implementation Details

We build ProtoGCN on the PYSKL [9] framework. The GCN backbone consists of 10 stages: the first stage maps 3 input channels to 96 base channels, followed by stages that inflate channel width at stages 5 and 8 (reaching 192 and 384 channels respectively) and downsample temporally at stages 5 and 8. The MTE ratio for mid-channels is 0.125 of the output channels. The PRN uses $n_{\text{pro}} = 100$ prototypes by default and a dropout rate of 0.1. The CSCL loss uses temperature $\tau_c = 0.125$, momentum $\mu = 0.9$, and a projection dimension $D = 256$. The loss weight is $\lambda = 0.1$.

We train all models for 65 epochs using SGD with momentum 0.9, weight decay 10^{-4}, and a cosine annealing learning rate schedule with initial rate 0.1. We use a batch size of 64 per GPU and train on 4 GPUs. For multi-stream ensembling, we combine predictions from joint, bone, joint motion, bone motion, joint 3D, and bone 3D modalities (6-stream ensemble).

4.3. Comparison with State-of-the-Art

Table 1 presents comparisons against existing methods on NTU RGB+D, NTU RGB+D 120, and Kinetics-Skeleton. With the 6-stream ensemble, ProtoGCN achieves **93.8%** X-Sub and **97.8%** X-View on NTU RGB+D, surpassing all prior methods. Compared to the previous best results on this dataset (JT-GraphFormer [43] at 93.4%/97.5%), ProtoGCN improves by +0.4 on X-Sub and +0.3 on X-View.

On the more challenging NTU RGB+D 120, ProtoGCN achieves **90.9%** X-Sub and **92.2%** X-Set, exceeding Block-GCN [45] (the previous best) by +0.6 on X-Sub and +0.7 on X-Set. The larger margins on NTU-120 are consistent with the hypothesis that prototype-based reconstruction is especially beneficial when more fine-grained discrimination is required across a larger class set.

On Kinetics-Skeleton, ProtoGCN achieves **51.9%** Top-1 and **75.6%** Top-5, improving upon DS-GCN [34] (50.6% Top-1) by +1.3. This result is particularly notable as Kinetics-Skeleton poses a large-scale, real-world recognition challenge where the benefits of fine-grained topology modeling are less obvious, demonstrating the broad applicability of our approach.

Even with the 2-stream ensemble, ProtoGCN (93.0%, 97.2%, 89.7%, 91.2%) is already competitive with many prior 6-stream methods, indicating that the single-stream representations learned by ProtoGCN are inherently more discriminative.

Table 2 shows results on FineGYM. ProtoGCN achieves **95.9%**, outperforming PoseConv3D [10] (94.3%) and PYSKL [9] (94.1%) by substantial margins. Critically, ProtoGCN surpasses even methods that use additional modalities (RGB flow), using only skeleton data, which underscores the effectiveness of our fine-grained topology learning.

4.4. Ablation Study

Component Contributions. Table 3 presents an ablation on the NTU-120 X-Sub setting, incrementally adding each component of ProtoGCN to a baseline GCN. The baseline achieves 87.8%. Adding MTE yields 88.1% (+0.3%), confirming that enriched topology signals from both intra- and

Table 1. Performance comparisons against state-of-the-art methods on the NTU RGB+D, NTU RGB+D 120, and Kinetics-Skeleton datasets in terms of classification accuracy (%).

Methods	Publication	NTU RGB+D		NTU RGB+D 120		Kinetics-Skeleton	
		X-Sub	X-View	X-Sub	X-Set	Top-1	Top-5
ST-GCN [37]	AAAI 2018	81.5	88.3	70.7	73.2	30.7	52.8
AS-GCN [17]	CVPR 2019	86.8	94.2	78.3	79.8	34.8	56.5
2s-AGCN [24]	CVPR 2019	88.5	95.1	82.5	84.2	36.1	58.7
MS-G3D [20]	CVPR 2020	91.5	96.2	86.9	88.4	38.0	60.9
DC-GCN+ADG [5]	ECCV 2020	90.8	96.6	86.5	88.1	-	-
MST-GCN [4]	AAAI 2021	91.5	96.6	87.5	88.8	38.1	60.8
CTR-GCN [3]	ICCV 2021	92.4	96.8	88.9	90.6	-	-
STF [14]	AAAI 2022	92.5	96.9	88.9	89.9	39.9	-
InfoGCN [6]	CVPR 2022	93.0	97.1	89.8	91.2	-	-
PYSKL [9]	ACM MM 2022	92.6	97.4	88.6	90.8	49.1	-
SkeletonGCL [12]	ICLR 2023	92.8	97.1	89.8	91.2	-	-
FR-Head [44]	CVPR 2023	92.8	96.8	89.5	90.9	-	-
GAP [33]	ICCV 2023	92.9	97.0	89.9	91.1	-	-
HD-GCN [16]	ICCV 2023	93.4	97.2	90.1	91.6	40.9	63.5
JT-GraphFormer [43]	AAAI 2024	93.4	97.5	89.9	91.7	-	-
DS-GCN [34]	AAAI 2024	93.1	97.5	89.2	91.1	50.6	-
BlockGCN [45]	CVPR 2024	93.1	97.0	90.3	91.5	-	-
ProtoGCN (2-ensemble)		93.0	97.2	89.7	91.2	49.9	74.0
ProtoGCN (4-ensemble)		93.5	97.5	90.4	91.9	51.3	75.1
ProtoGCN (6-ensemble)		**93.8**	**97.8**	**90.9**	**92.2**	**51.9**	**75.6**

Table 2. Performance comparisons against state-of-the-art methods on FineGYM in terms of classification accuracy (%).

Methods	Modality	FineGYM
I3D [2]	RGB	64.4
TSN [32]	RGB+Flow	79.8
TSM [18]	RGB+Flow	81.2
LT-S3D [35]	RGB	88.9
SkeletonMAE [36]	Skeleton	91.8
PYSKL [9]	Skeleton	94.1
PoseConv3D [10]	Skeleton+Limb	94.3
ProtoGCN (Ours)	Skeleton	**95.9**

Table 3. Ablation study on the contribution of each component in ProtoGCN under the NTU-120 X-Sub setting.

Baseline	MTE	PRN	CSCL	Acc (%)
✓	–	–	–	87.8
✓	✓	–	–	88.1 (↑ 0.3)
✓	✓	–	✓	88.3 (↑ 0.5)
✓	✓	✓	–	88.5 (↑ 0.7)
✓	✓	✓	✓	**89.0 (↑ 1.2)**

inter-sample graphs improve representation quality. Adding CSCL on top of MTE gives 88.3% (+0.5% total), showing that class-specific contrastive regularization further sharpens the features. Adding PRN while removing CSCL yields 88.5% (+0.7%); the largest individual gain from any single additional component, highlighting the importance of prototype-based reconstruction. The full ProtoGCN with all three components achieves 89.0% (+1.2%), demonstrating that the components are complementary and jointly contribute to performance.

Comparison with Graph Contrastive Baselines. Table 4 (left) compares ProtoGCN against alternative graph contrastive learning methods applied to the same baseline. SkeletonGCL [12] achieves 88.0% and FR-Head [44] achieves 88.2%, while ProtoGCN reaches 89.0%. This confirms that the improvements from ProtoGCN are not simply due to contrastive learning in general, but specifically due to the combination of topology-aware prototype reconstruction and class-specific contrast.

Table 4 (right) ablates the topology components of MTE. Removing $\mathbf{A}_{\text{inter}}$ causes a larger drop (88.5%) than removing $\mathbf{A}_{\text{intra}}$ (88.8%), and removing both gives 88.4%, suggesting that the inter-sample difference topology provides the most informative signal for distinguishing between similar actions.

Table 4. Comparison of classification accuracies (%) based on (*left*) existing graph contrastive methods and (*right*) different enhanced topologies for the network.

Methods	Acc (%)	Methods	Acc (%)
Baseline	87.8	ProtoGCN	**89.0**
w/ SkeletonGCL [12]	88.0	w/o $\mathbf{A}_{\mathbf{intra}}$	88.8
w/ FR-Head [44]	88.2	w/o $\mathbf{A}_{\mathbf{inter}}$	88.5
Ours	**89.0**	w/o $\mathbf{A}_{\mathbf{intra}}, \mathbf{A}_{\mathbf{inter}}$	88.4

Table 5. Average performance (%) on different difficulty level classes sorted by accuracy under the NTU-120 X-Sub setting.

Difficulty Level	Baseline	ProtoGCN	Δ
class 1-10	64.1	67.1	**+3.0**
class 11-30	78.6	81.1	**+2.5**
class 31-60	88.6	89.7	**+1.1**
class 61-120	96.0	96.4	**+0.4**

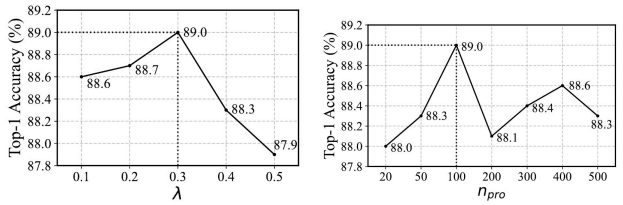

Figure 2. Ablation study on the influences of weight λ (left) and the memory capacity n_{pro} (right) under the NTU-120 X-Sub setting.

Analysis by Action Difficulty. Table 5 analyzes performance across action classes grouped by difficulty (sorted by baseline accuracy, where class 1–10 are hardest). ProtoGCN provides the largest gains on the hardest classes: +3.0% on classes 1–10 and +2.5% on classes 11–30, with progressively smaller gains on easier classes (+1.1% and +0.4% for classes 31–60 and 61–120). This demonstrates that the prototype-based reconstruction specifically benefits the most challenging, fine-grained discrimination scenarios.

Hyperparameter Sensitivity. Figure 2 shows the sensitivity of ProtoGCN to the loss weight λ and prototype count n_{pro} on NTU-120 X-Sub. Performance peaks around $\lambda = 0.1$; very small values underutilize the contrastive signal while large values over-regularize the topology features. For n_{pro}, performance improves from small values up to around 100, then plateaus; 100 prototypes provides sufficient capacity to cover the space of motion patterns without over-parameterizing the memory. We adopt $\lambda = 0.1$ and $n_{\text{pro}} = 100$ as our defaults.

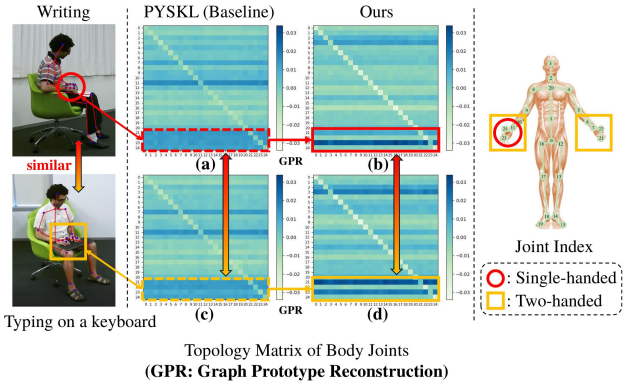

Figure 3. Illustration of the skeletons and learned topologies for similar actions *Writing* and *Typing on a Keyboard* (deeper color indicates stronger relationships between corresponding joints). ProtoGCN (right) learns more discriminative, action-specific joint relationships compared to the baseline (left).

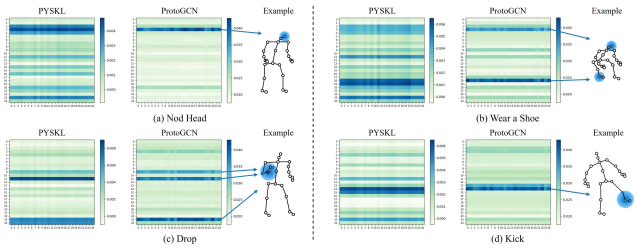

Figure 4. Visualization of the topologies learned by PYSKL [9] and ProtoGCN across four actions. ProtoGCN produces action-specific topologies that emphasize the most discriminative joint relationships.

4.5. Qualitative Analysis

Learned Topologies. Figure 3 illustrates the topologies learned by PYSKL and ProtoGCN for "Writing" and "Typing on a Keyboard." PYSKL learns similar distributed topologies for both actions, making them difficult to distinguish. ProtoGCN produces topologies that are visibly differentiated: for Writing, the hand and wrist joints exhibit strong connections reflecting the fine pen-control motions, while for Typing, the finger joints show more symmetric bilateral patterns. This qualitative difference validates that MTE and PRN successfully highlight the specific joint relationships that distinguish the two actions.

Figure 4 extends this analysis to four additional actions, showing that ProtoGCN consistently learns more action-specific topologies compared to the PYSKL baseline, with deeper (stronger) connections in the body regions most relevant to each action.

Per-Class Accuracy Improvement. Figure 5 shows action classes where ProtoGCN improves over PYSKL by more than 1% on NTU-120. The improvements are concentrated

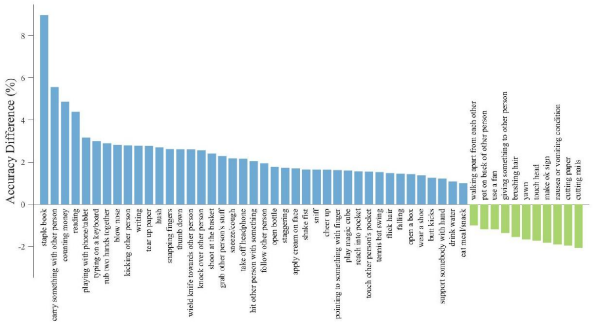

Figure 5. Action classes with accuracy difference higher than 1% between our method and PYSKL [9] on the NTU-120 dataset.

in actions involving subtle hand and wrist movements (writing, typing, phone use) and fine-grained lower body patterns, consistent with the hypothesis that prototype-based topology reconstruction specifically benefits actions distinguished by localized joint dynamics.

5. Conclusion

We have presented ProtoGCN, a novel framework for skeleton-based action recognition that addresses the fundamental challenge of distinguishing similar actions through a prototypical perspective on motion topology. ProtoGCN integrates three complementary innovations: Motion Topology Enhancement (MTE), which enriches GCN layers with intra-sample and inter-sample topology signals to capture fine-grained joint-motion differences; a Prototype Reconstruction Network (PRN), which reconstructs topology features from a learnable prototype memory to expose discriminative motion patterns; and Class-Specific Contrastive Learning (CSCL), which enforces class-level compactness and dispersion using momentum-updated class centers. Extensive experiments across four benchmarks demonstrate that ProtoGCN achieves new state-of-the-art results on NTU RGB+D 60, NTU RGB+D 120, Kinetics-Skeleton, and FineGYM. Analysis by action difficulty confirms that the gains are strongest for the hardest fine-grained classes, validating our core hypothesis. Qualitative topology visualizations show that ProtoGCN consistently learns more discriminative, action-specific joint-relationship patterns compared to prior methods. We believe the prototypical perspective introduced here opens promising directions for future work, including hierarchical prototype memories for multi-granularity action modeling and dynamic prototype updates conditioned on temporal context.

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